

Acceptance requires exactly one compatible candidate.

OUTCOME	ROWS	DISTINCT IDS
Matched	61	60
All candidates contradicted	40	40
Ambiguous	10	10
Conflicting source data	1	1
No candidate in reference	0	0

112 worksheet rows · 111 distinct vehicle IDs.
Unresolved vehicles stay in the source denominator and never enter the VIO sheet.

DECISION POLICY IN THIS RUN

Version criteria are capacity, fuel and power; variant criteria are body, engine and drive. Missing evidence is unknown and never becomes agreement.

A documented manufacturer model code decoded from a scoped identifier can make a sole compatible candidate sufficient; a documented disagreement is a veto. Extra agreements never eliminate a compatible sibling.

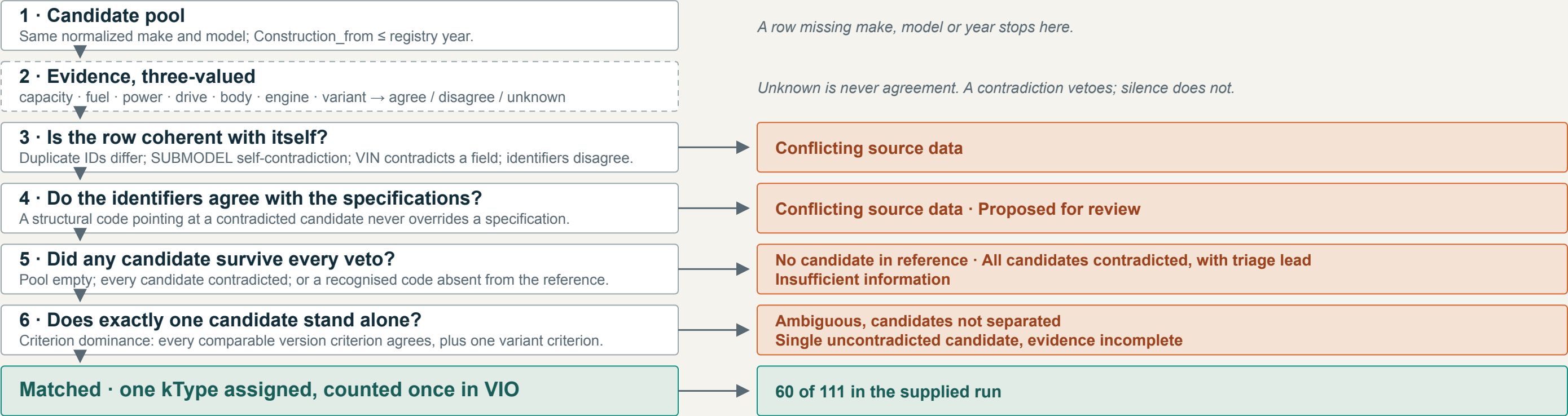
Year mode is **gate**: a candidate whose `Construction_from` year is later than the registry year is excluded.
`Construction_to` does not gate, because the NZ year may be first registration.

Exact power agreement · 0 tolerance used

The one-kilowatt band orders review only.
No acceptance in this run depends on it.

A row stops at the first thing wrong with it.

Seven stages, in this order. Six of them can only stop a row; one can assign a kType.



The order is load-bearing: every stage returns, so a row is reported against the first obstacle it meets rather than the last. The default policy reads no weights, no threshold and no score.

Each unresolved group names a field and both values.

GROUP	FIELD	SOURCE VS REFERENCE	OUTCOME	IDS
X-TRAIL ST 2.5P/6CVT	power	125 kW vs 126 kW (kTypes 107383, 107384); 27 of 28 candidates	All candidates contradicted	20
RANGER WILDTRAK 2.0D/4WD	capacity; power	1996 cc vs 2184–3198 cc; 157 kW vs 57–147 kW across 13 candidates	All candidates contradicted	14
C-CLASS C 200 1.5P/9AT	capacity	Power, fuel, variant and engine agree with kType 137964; capacity alone disagrees, 1491 cc against 1991 cc	All candidates contradicted	6
COLORADO LS RC SC 2.8D/4WD	—	kTypes 100868 and 100871 both remain compatible	Ambiguous	10
C-CLASS C 200 1.5P/9AT	capacity	Same, plus the VIN <i>Grundbaumuster</i> 205380 authoritatively confirms 137964 against the source's own capacity	Conflicting source data	1

Three different problems: a publisher disagreement, an absent catalogue version, a registry recording error. Naming which is which is what a reviewer acts on.

The rule I deleted, and why.

This is the v2 decision that cost nine assignments, recorded in the shipped rule table.

Removed from the body table

NZ STATION WAGON → Estate / SUV / Closed Off-Road Vehicle

NZ SPORTS CAR → Coupe / Convertible / Saloon

NZ UTILITY → Pickup / Platform/Chassis

Judgement, not vocabulary

No publisher states these correspondences. The two taxonomies are independent, so the table was asserting my opinion with the authority of a reference.

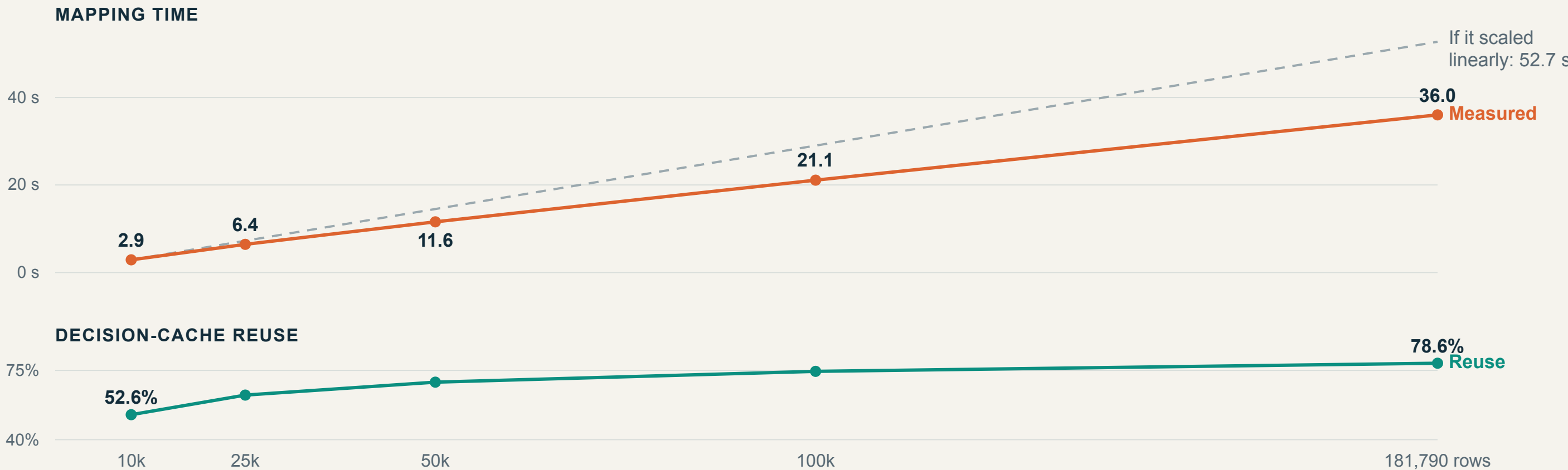
What survives is name identity only. A registration body pairs with a reference structure **solely where both publishers print the same word**: NZTA VIRM Table 2-2-8 names SL "Saloon" and CV "Convertible"; TecDoc KT 086 states "Saloon" and "Convertible" in the supplied extract. A reviewer can check both sides. Nothing else is paired.

Covers 2 of NZTA’s 26 body types. The rule file declares that incompleteness itself, so a gap is never read as evidence: in this run the body field agrees or stays silent, and has never once contradicted a candidate.

The cost is visible in the coverage figure and nowhere hidden: nine assignments, from 51 to 42, before sourced identifier evidence recovered past it. That is the trade this project makes everywhere.

Cost grows more slowly than volume.

The published 2020 New Zealand register, not supplied with the task: 36 seconds to map, under a minute to a complete result set as CSV. Eighteen times the rows cost twelve times the time.



Source rows, linear scale. Median of three runs per point; run-to-run spread at most 4.6%. The published file is unordered, so each prefix is representative.

The matcher is not the bottleneck. Workbook serialization is.

PHASE · 181,790 ROWS · ROUNDED	CSV OUT	WITH XLSX
Load and normalize source	1.6 s	1.5 s
Map and stream evidence	36.2 s	35.9 s
Build sheets	7.5 s	7.5 s
Write sheets as CSV	10.1 s	9.5 s
Write workbook (XLSX)	—	430.0 s
Run metadata (hashing)	0.2 s	0.2 s
Write performance report	3.7 s	3.5 s
Total	59.3 s	488.0 s

WHAT THIS SAYS TO DO

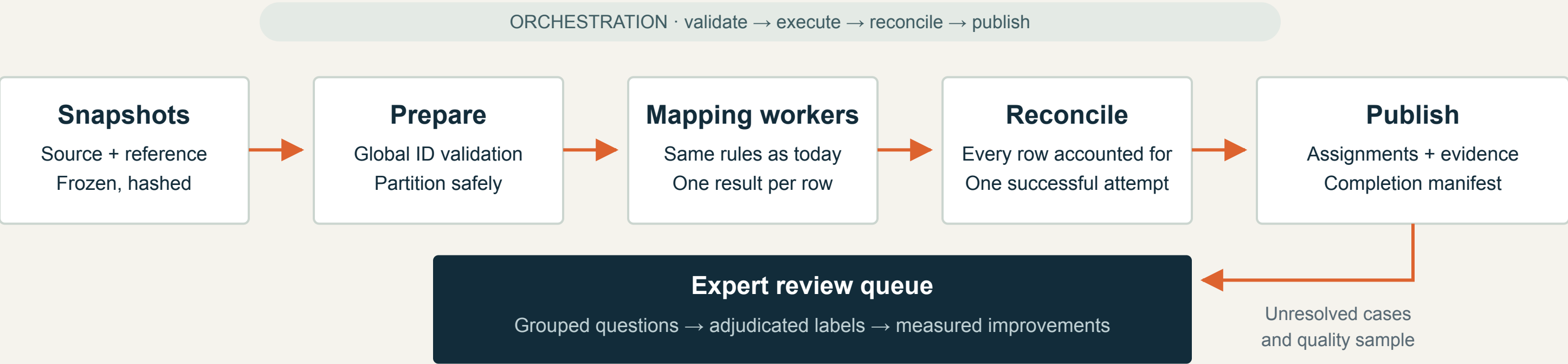
XLSX serialization is 88% of wall clock and 45× the cost of the identical data as CSV, because Results and Review_Detail carry ~360,000 wide rows.

The fix is the container, not the code: route row-level sheets to CSV above a threshold, exactly as candidate evidence already is, and keep the workbook for the sheets a person reads. That returns a full run to about a minute.

Not done, deliberately. At the 111-row scale of this submission the workbook takes about five seconds and is the right format.

Scale the work. Keep the decisions traceable.

A sketch for the costed expansion plan, if and only if the pilot passes all three gates.



Retries: separate attempt outputs **Publication:** manifest written after reconciliation **Adjudication:** the only source of quality labels

Adding workers does not resolve oversized candidate groups. Those need a validated additional retrieval or comparison pass.

Reviewer corrections become rules.

That is where the learning belongs. Each stage produces what the next one needs, and a model never assigns a kType.

STAGE 0 · THE PILOT

Adjudicate

Expert decisions on a real batch. This is the measurement, and it produces the first labelled data this problem has ever had.

Gate: precision meets the agreed floor and effort falls.

STAGE 1 · CLASSICAL ML

Induce rules

Mine adjudications for readable conditions (*where BODY_TYPE is UTILITY and the cab code is X, reviewers chose Pickup*), each carrying its support count and confidence. Proposed to the owner, not applied.

Plus a learned ranker for review order. No language model.

STAGE 2 · BEDROCK

Explain exceptions

Natural-language explanation of one conflict, from evidence already computed. Measured on reviewer effort, never on accuracy.

Stage 3, a learned scorer, only if 1 and 2 pay.

Per row and high volume → rules or a small classical model. Per exception and read by a human → a language model. A per-row LLM call across 181,790 vehicles is indefensible; across 2,600 exceptions it is bounded.

Sixty accepted rows are nine decisions.

Identical evidence reuses one cached decision, so rows sharing it prove nothing extra. Sample size has to be counted in decisions.

DECISIONS ADJUDICATED, NO ERRORS	PRECISION FLOOR IT SUPPORTS
9 · the whole accepted set today	0.72
30	0.91
50	0.94
100	0.97
150	0.98
300	0.99

One-sided 95% Clopper–Pearson lower bound, zero errors. Counted in distinct decisions, not rows, and the batch must hold that many.

WHAT THIS CHANGES ABOUT THE ASK

Sampling **by decision** costs a fraction of sampling by row, because a few configurations dominate volume: the commonest decision covers 835 vehicles, while 237 of 483 appear once.

TARGET	BY ROW	BY DECISION
150 decisions	~1,405 rows	150

Count your batch first. One model year holds 483 distinct decisions and vintages share none, so the reachable floor is a property of the batch.